

## Question 1

- Determine the motion of a mass-spring-damper system if parameter values are  $m = 2$ ,  $b = 6$  and  $k = 4$ , and the mass (initially at rest) is released from the position  $x = 1$ .

System:  $m\ddot{x} + b\dot{x} + kx = 0$   
 $2\ddot{x} + 6\dot{x} + 4x = 0$

characteristic equation  $2\lambda^2 + 6\lambda + 4 = 0$

The roots are:  $\lambda = \frac{-6 \pm \sqrt{36 - 32}}{4} = \frac{-6 \pm \sqrt{4}}{4} = \frac{-6 \pm 2}{4} = -1 \text{ \& } -2$

two distinct  
real roots

The general solution:  $x(t) = C_1 e^{-t} + C_2 e^{-2t}$

To determine  $C_1$  &  $C_2$ , we will use initial conditions

$x(t=0) = C_1 + C_2 = 1$  (initial position)

$\dot{x}(t) = \frac{dx(t)}{dt} = -C_1 e^{-t} - 2C_2 e^{-2t}$

$\dot{x}(t=0) = -C_1 - 2C_2 = 0$  (at rest)

$$\begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \frac{1}{-2+1} \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{-1} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$



$$x(t) = 2e^{-t} - e^{-2t}$$

## Question 2

- Determine the motion of a mass-spring-damper system if parameter values are  $m = 1$ ,  $b = 2$  and  $k = 1$ , and the mass (initially at rest) is released from the position  $x = 4$ .

System  $m\ddot{x} + b\dot{x} + kx = 0$   
 $\ddot{x} + 2\dot{x} + x = 0$

Characteristic equation:  $\lambda^2 + 2\lambda + 1 = 0$

The roots are:  $\lambda = \frac{-2 \pm \sqrt{4-4}}{2} = \underline{\underline{-1, -1}}$  two same roots!

The general solution is:

$$x(t) = [C_1 + C_2 t] e^{\lambda t}$$

$\lambda = -1$

To calculate  $C_1$  &  $C_2$ , use initial conditions.

$$x(t=0) = C_1 = 4$$

$$\dot{x} = \frac{dx}{dt} = -(C_1 + C_2 t)e^{-t} + C_2 e^{-t}$$

$$\dot{x}(t=0) = -\underbrace{C_1}_4 + C_2 = 0$$

$$\Rightarrow C_2 = 4$$

Final answer:

$$x(t) = [4 + 4t] e^{-t}$$

## Question 3

- Determine the motion of a mass-spring-damper system if parameter values are  $m = 1$ ,  $b = 4$  and  $k = 5$ , and the mass (initially at rest) is released from the position  $x = 2$ .

System:  $m\ddot{x} + b\dot{x} + kx = 0$   
 $\ddot{x} + 4\dot{x} + 5x = 0$

Characteristic equation:  $\lambda^2 + 4\lambda + 5 = 0$

The roots are:  $\lambda = \frac{-4 \pm \sqrt{16-20}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = \frac{-4 \pm \sqrt{4}j}{2}$   
 $= \underline{\underline{-2 \pm j}}$

$$= -2 \pm j$$

The general equation is:  $x(t) = C_1 e^{-2t} \cos t + C_2 e^{-2t} \sin t$

Use initial conditions to find  $C_1$  &  $C_2$

$$x(t=0) = C_1 = 2$$

$$\dot{x} = \frac{dx}{dt} = -2C_1 e^{-2t} \cos t - C_1 e^{-2t} \sin t + -2C_2 e^{-2t} \sin t + C_2 e^{-2t} \cos t$$

$$\dot{x}(t=0) = -2 \underbrace{C_1}_2 + C_2 = 0$$

$$\Rightarrow C_2 = 4$$

Final answer:  $x(t) = 2e^{-2t} \cos t + 4e^{-2t} \sin t$

## Question 4

- Give the nonlinear control equations for the system:

$$(2\sqrt{\theta} + 1)\ddot{\theta} + 3\dot{\theta}^2 - \sin(\theta) = \tau$$

- Choose gains so that this system is always critically damped with closed-loop stiffness of 10.

System:  $(2\sqrt{\theta} + 1)\ddot{\theta} + 3\dot{\theta}^2 - \sin(\theta) = \tau$

Control step 1. Model-based compensation.  $\tau = (2\sqrt{\theta} + 1)\ddot{f} + 3\dot{\theta}^2 - \sin(\theta)$

Control step 2:  $f = \ddot{\theta}_{ref} - k_d(\dot{\theta} - \dot{\theta}_{ref}) - k_p(\theta - \theta_{ref})$

$$\text{set } k_p = k_{cl} = \underline{\underline{10}}$$

$$\text{let } k_d = 2\sqrt{k_p} = \underline{\underline{2\sqrt{10}}}$$

## Question 5

- Design a trajectory-following control system for a system with the following dynamic equations:

$$\begin{aligned} m_1 l_1^2 \ddot{\theta}_1 + m_1 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 &= \tau_1 \\ m_2 l_2^2 (\ddot{\theta}_1 + \ddot{\theta}_2) + v_2 \dot{\theta}_2 &= \tau_2 \end{aligned}$$

- Additional question: What is wrong with robot model?

Write in  $M, V, G$  form

$$\begin{bmatrix} m_1 l_1^2 & 0 \\ m_2 l_2^2 & m_2 l_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} m_1 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \\ v_2 \dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

Control step 1: 
$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} m_1 l_1^2 & 0 \\ m_2 l_2^2 & m_2 l_2^2 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} + \begin{bmatrix} m_1 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \\ v_2 \dot{\theta}_2 \end{bmatrix}$$

Control step 2: Design 
$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} \ddot{\theta}_{1, \text{ref}} \\ \ddot{\theta}_{2, \text{ref}} \end{bmatrix} + \begin{bmatrix} k_{d1} & 0 \\ 0 & k_{d2} \end{bmatrix} \begin{bmatrix} \dot{\theta}_{1, \text{ref}} - \dot{\theta}_1 \\ \dot{\theta}_{2, \text{ref}} - \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} k_{p1} & 0 \\ 0 & k_{p2} \end{bmatrix} \begin{bmatrix} \theta_{1, \text{ref}} - \theta_1 \\ \theta_{2, \text{ref}} - \theta_2 \end{bmatrix}$$

Set  $k_{p1}, k_{p2}$  for required stiffness

$$\left. \begin{aligned} k_{d1} &= 2\sqrt{k_{p1}} \\ k_{d2} &= 2\sqrt{k_{p2}} \end{aligned} \right\} \text{for critical damping}$$

Additional question: The  $M$ -matrix is not symmetrical !!